

Our understanding of the structure of the atom has evolved over centuries and continues to change in the light of new evidence. When the existence of atoms was proposed by John Dalton in 1808, it resurrected an idea that originated around two thousand years earlier in ancient Greece and India. The philosophy that matter was composed of tiny, solid spheres that could not be divided into smaller parts has developed into an evidence-based theory that allows us to predict and explain the properties and behaviour of matter.

Science as a Human Endeavour

- Findings from a range of scientific experiments contributed to the understanding of the atom, enabling scientists, including Dalton, Thomson, Rutherford, Bohr and Chadwick to develop models of atomic structure and make reliable predictions about the mass, charge and location of the sub-atomic particles.

Science understanding

- the relative atomic mass (atomic weight), A_r , is the ratio of the average mass of the atom to $1/12$ the mass of an atom of ^{12}C ; relative atomic masses of the elements are calculated from their isotopic composition
- isotopes are atoms of an element with the same number of protons but different numbers of neutrons and are represented in the form ^AX (IUPAC) or X-A
- isotopes of an element have the same electron configuration and possess similar chemical properties but have different physical properties
- mass spectrometry involves the ionisation of substances and the separation and detection of the resulting ions; the spectra which are generated can be analysed to determine the isotopic composition of elements and interpreted to determine relative atomic mass

2.1 Atomic theory



FIGURE 2.1.1 John Dalton (1766–1844) proposed that matter was composed of atoms.

Around 450 BCE, the Greek philosopher Leucippus proposed that, if a piece of matter were subdivided over and over again, eventually it would not be able to be divided any further. A follower of Leucippus called Democritus expanded on this idea and named the final piece of matter *atomos*, meaning ‘indivisible’. He believed different substances had atomos of different sizes and shapes, which gave substances different properties. Unfortunately, Democritus’ ideas were not widely accepted; the preferred teachings were those of Aristotle, who believed that all matter was composed of the four elements: earth, air, fire and water.

The notion of atomos was not revisited until 1808, when English chemist John Dalton (Figure 2.1.1) published *A New System of Chemical Philosophy*, in which he explained his observations of chemical reactions in terms of tiny particles that he called ‘atoms’. Dalton thought that all matter was made up of these indivisible atoms and that the atoms of different elements had different weights. He further proposed that compounds are formed from the combination of the atoms of two or more elements in simple numerical ratios. He stated that atoms of a given element are identical in size and mass and when atoms combine to form compounds, they do so only in the simplest ratio. Not all of these points still hold true, however, his innovative ideas became the foundation of modern atomic theory.

THE STRUCTURE OF ATOMS

Most of the points of Dalton’s atomic theory were well received and could be supported with experimental data, but by the end of the 1800s, there was increasing evidence to suggest that atoms are made up of smaller, subatomic particles. This shifted the focus of scientists of the time from the existence of atoms to the subatomic composition and structure of atoms.

One such scientist was Joseph John Thomson (Figure 2.1.2) who carried out experiments with an apparatus known as a cathode ray tube, which would later be utilised in the first televisions. A cathode ray tube consists of a sealed glass tube with metal terminals at each end that act as electrodes. When electricity passes through the tube, a stream of particles is emitted and the tube glows with a coloured light.

In 1897, Thomson discovered that, regardless of the gas in the tube or the type of metal electrodes used, the particles flow from the negative electrode to the positive electrode and the stream of particles is repelled by another negative electrode. He concluded that the particles must be negatively charged and be present in all atoms. Since atoms are electrically neutral, he also proposed that they must contain positive particles as well so that together the charges cancel out.

To explain the existence of these two subatomic particles, Thomson developed the ‘plum pudding’ model of the atom where the negatively charged particles were spread throughout the positively charged sphere of the atom like the raisins of a plum pudding. These negatively charged particles became known as **electrons**.

As scientists gathered more experimental evidence, it became apparent that Thomson’s model of the atom was not correct. In 1910, New Zealand-born Ernest Rutherford (Figure 2.1.3) led a group of scientists who performed an experiment that changed the accepted view of the atom.

Rutherford fired a beam of positively charged alpha particles at a very thin sheet of gold foil. Most of the alpha particles passed straight through the foil and produced flashes on a fluorescent screen on the other side. Rutherford was surprised to discover, however, that some particles were deflected as they passed through the foil, and some even bounced back. This experiment is described in Figure 2.1.4.

Rutherford concluded that most of an atom must be empty space to allow the majority of the alpha particles to pass through it. Most of the mass of an atom and its positive charge must be located in a tiny, central region. This region must have deflected the alpha particles. He called this region the **nucleus** of the atom and surrounded it with the negative electrons similar to planets orbiting the Sun.



FIGURE 2.1.2 British physicist J. J. Thomson

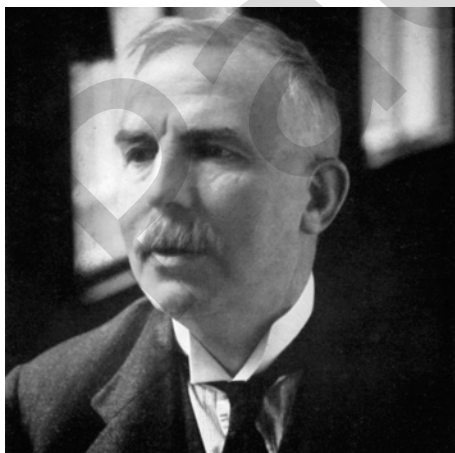


FIGURE 2.1.3 New Zealand-born physicist Ernest Rutherford

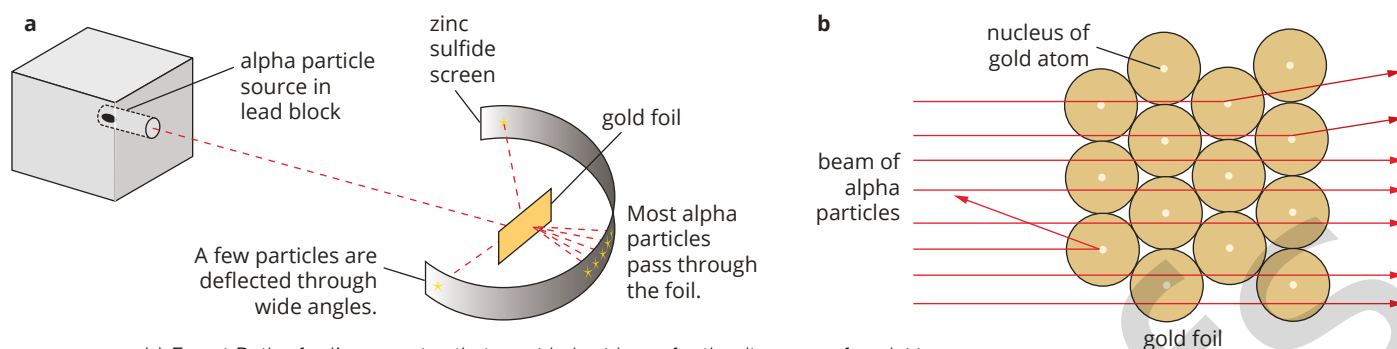


FIGURE 2.1.4 (a) Ernest Rutherford's apparatus that provided evidence for the discovery of nuclei in atoms. (b) Only those alpha particles that closely approach the nuclei in the gold foil are deflected significantly. Most particles pass almost directly through the foil.

Rutherford's model of orbiting electrons would be further improved by Niels Bohr. You will learn more about the Bohr model of the atom in Chapter 3.

In further experiments by Rutherford in 1920, he found that firing a stream of alpha particles into nitrogen gas produced a signal typical of hydrogen nuclei. He went on to determine that there are hydrogen nuclei in all other nuclei. He called this elementary particle present in all nuclei, the **proton**.

Rutherford also had reason to believe the nucleus contained a yet undiscovered neutral particle. He believed it to be a proton with a closely orbiting electron, as his experimental data indicated that a neutral particle with a mass similar to a proton should exist.

A previous student, then colleague of Rutherford, the English physicist James Chadwick later confirmed the existence of Rutherford's proposed neutral particle. Chadwick showed it is a distinct particle and not the proton–electron pairing previously suggested. He called this new elementary particle, the **neutron**.

Electrons

Electrons are negatively charged particles which behave like a cloud of negative charge around the nucleus. This cloud gives the atom its size and volume. An electron has a mass approximately 1800 times smaller than that of a proton or neutron. Therefore, electrons contribute very little to the total mass of an atom. However, the volume of space occupied by the cloud of electrons is 10 000–100 000 times larger than that of the nucleus.

Negative particles attract positive particles. This is called **electrostatic attraction**. The electrostatic attraction of the atom's electrons to the protons within the nucleus keeps the atom bound together. The charge on an electron is equal but opposite to the charge on a proton. Electrons are said to have a charge of -1 whereas protons have a charge of $+1$.

In some circumstances, electrons can be easily removed from an atom. For example, when you rub a rubber balloon on a woollen jumper or dry hair, electrons are transferred to the balloon and the balloon develops a negative charge. The negative charge is observed as an electrostatic force that can stick the balloon to a wall. You will learn more about the removal of electrons from atoms when looking at ionic compounds in Chapter 5.

The electricity that powers lights and appliances is the result of electrons moving in a current through wires. Sparks and lightning are also caused by electrons moving through air (Figure 2.1.5).



FIGURE 2.1.5 Lightning is a sudden electrostatic discharge that occurs during a thunderstorm.

The nucleus

The nucleus of an atom is approximately 10 000–100 000 times smaller than the size of the atom. To put this in perspective, if an atom were the size of Perth Stadium, then the nucleus would be about the size of a pea (Figure 2.1.6). Nonetheless, the nucleus contributes around 99.97% of the atom's mass. This means that atomic nuclei are extremely dense.

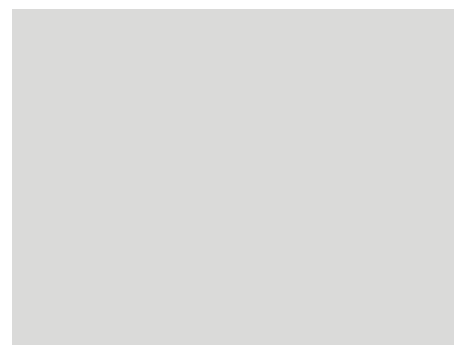


FIGURE 2.1.6 If an atom were the size of Perth stadium (built in 2017), then the nucleus would be about the size of a pea.

The subatomic particles in the nucleus, the protons and neutrons, are referred to collectively as **nucleons**. Protons are positively charged particles with a mass of approximately 1.673×10^{-27} kg. Neutrons are almost identical in mass to protons.

Table 2.1.1 summarises the properties of protons, neutrons and electrons.

TABLE 2.1.1 Properties of the subatomic particles

Particle	Symbol	Charge	Mass relative to a proton	Mass (kg)
proton	p^+	+1	1	1.673×10^{-27}
neutron	n^0	0	1	1.675×10^{-27}
electron	e^-	-1	$\frac{1}{1800}$	9.109×10^{-31}

EXTENSION

Quarks

The proton, neutron and electrons are not the only particles that have been discovered and characterised. Towards the early 1960s there were so many particles discovered that they were collectively referred to as the 'particle zoo'. These were all believed to be elementary particles, that is, particles which are not in turn made of other particles.

In the early 1960s two physicists, Murray Gell-Man and George Zweig, separately suggested these particles, including the protons and neutrons, weren't elementary particles but were composed of a subset of particles known as quarks. The six types, or flavours, of quarks are up, down, strange, charm, top and bottom.

The quarks are now included in a list of 17 elementary particles with the most recently discovered, the Higgs Boson, confirmed in 2012.

2.1 Review

SUMMARY

- All substances are made up of atoms.
- Atoms are composed of a small, positively charged nucleus surrounded by a negatively charged cloud of electrons.
- The nucleus is made up of two subatomic particles—protons and neutrons. These particles are referred to as nucleons.
- Protons and neutrons are similar in size and mass while electrons are approximately 1800 times smaller.
- The charges on protons and electrons are equal but opposite.
- Neutrons are uncharged.
- Positive and negative particles experience an attractive force known as electrostatic attraction.

KEY QUESTIONS

- 1 Name the three subatomic particles that atoms are composed of.
- 2 What subatomic particles make up the most mass of an atom and where are they found?
- 3 How are electrons held within the cloud surrounding the nucleus?

2.2 Describing atoms

Even before the structure of the atom was well understood, scientists had begun classifying and categorising them using qualitative and quantitative data obtained through experimentation. Thanks to the contributions of Dalton and other scientists, objective measurements of the composition of atoms can now be made and they can be described using a number of derived quantities.

DESCRIBING ATOMS

Fundamentally, the characteristics of atoms are derived from the numbers of the subatomic particles they contain – the proton, the neutron and the electron.

Elements

As Dalton predicted, elements are made of just one type of atom and these atoms are identical. Most non-metallic elements, such as sulfur, form molecules with a definite number of sulfur atoms (8), however, some non-metal such as carbon, can form extensive networks numbering in the thousands of atoms (Figure 2.2.1). So far, scientists have discovered 118 different elements. About 98 of these occur in nature, the other elements have only been observed in the laboratory.

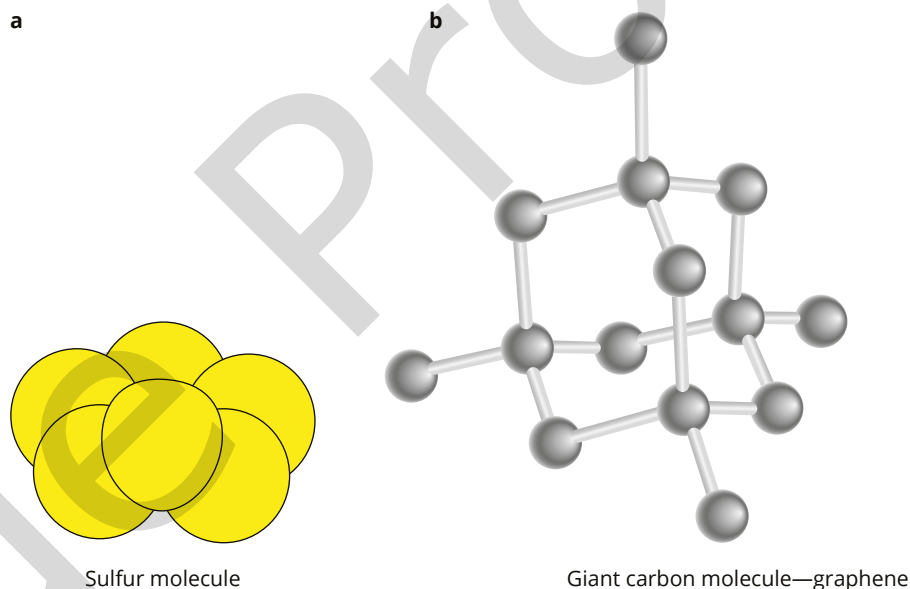


FIGURE 2.2.1 (a) Most non-metal elements, such as sulfur, form molecules. (b) Some other elements, such as carbon, form covalent network lattices or giant molecules given by the example here of graphene.

Each element has a unique name and **chemical symbol**. Table 2.2.1 lists the chemical symbols of some of the most common elements.

TABLE 2.2.1 Chemical symbols and names of some of the most common elements

Element	Symbol	Element	Symbol
aluminium	Al	mercury	Hg
argon	Ar	nitrogen	N
carbon	C	oxygen	O
chlorine	Cl	potassium	K
copper	Cu	silver	Ag
hydrogen	H	sodium	Na
iron	Fe	uranium	U

The chemical symbol is usually made up of one or two letters. The first letter is always capitalised and subsequent letters are always lower case.

In many cases, the chemical symbol corresponds to the name of the element. For example, nitrogen has the chemical symbol N, chlorine has the chemical symbol Cl and uranium has the chemical symbol U.

However, some chemical symbols do not correspond to the name of the element. For example, sodium has the chemical symbol Na, potassium has the chemical symbol K and iron has the chemical symbol Fe. This is because the chemical symbols have been derived from the Latin or Greek names of the elements. In Latin, sodium is known as *natrium*, potassium, *kalium* and iron, *ferrum*.

The atomic symbols are usually displayed in a periodic table as shown in Figure 2.2.2. The **periodic table** helps to group elements that have similar chemical properties. You will learn more about the periodic table in Chapter 3.

1 H hydrogen																	2 He helium						
3 Li lithium	4 Be beryllium																	5 B boron	6 C carbon	7 N nitrogen	8 O oxygen	9 F fluorine	10 Ne neon
11 Na sodium	12 Mg magnesium																	13 Al aluminium	14 Si silicon	15 P phosphorus	16 S sulfur	17 Cl chlorine	18 Ar argon
19 K potassium	20 Ca calcium	21 Sc scandium	22 Ti titanium	23 V vanadium	24 Cr chromium	25 Mn manganese	26 Fe iron	27 Co cobalt	28 Ni nickel	29 Cu copper	30 Zn zinc	31 Ga gallium	32 Ge germanium	33 As arsenic	34 Se selenium	35 Br bromine	36 Kr krypton						
37 Rb rubidium	38 Sr strontium	39 Y yttrium	40 Zr zirconium	41 Nb niobium	42 Mo molybdenum	43 Tc technetium	44 Ru ruthenium	45 Rh rhodium	46 Pd palladium	47 Ag silver	48 Cd cadmium	49 In indium	50 Sn tin	51 Sb antimony	52 Te tellurium	53 I iodine	54 Xe xenon						
55 Cs caesium	56 Ba barium	57–71 lanthanoids	72 Hf hafnium	73 Ta tantalum	74 W tungsten	75 Re rhenium	76 Os osmium	77 Ir iridium	78 Pt platinum	79 Au gold	80 Hg mercury	81 Tl thallium	82 Pb lead	83 Bi bismuth	84 Po polonium	85 At astatine	86 Rn radon						
87 Fr franchium	88 Ra radium	89–103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganeson						
Lanthanide series			57 La lanthanum	58 Ce cerium	59 Pr praseodymium	60 Nd neodymium	61 Pm promethium	62 Sm samarium	63 Eu europium	64 Gd gadolinium	65 Tb terbium	66 Dy dysprosium	67 Ho holmium	68 Er erbium	69 Tm thulium	70 Yb ytterbium	71 Lu lutetium						
Actinide series			89 Ac actinium	90 Th thorium	91 Pa protactinium	92 U uranium	93 Np neptunium	94 Pu plutonium	95 Am americum	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium						

KEY

Non-metals

Metals

Metalloids

atomic number

symbol

name

13

Al

aluminium

FIGURE 2.2.2 The periodic table groups the chemical elements according to their atomic structure.

CHEMFILE

Viewing atoms

Dalton's atomic theory of matter assumed that atoms are spherical. However, atoms cannot be seen with conventional microscopes. Therefore, there was no way to confirm the shape of atoms. It wasn't until 1981 that a microscope capable of viewing atoms was developed by IBM researchers Gerd Binnig and Heinrich Rohrer. This type of microscope is known as a **scanning tunnelling microscope (STM)**. Using STMs, scientists confirmed that atoms are indeed spherical.

STMs use an extremely sharp metal tip to detect atoms. The tip is scanned, line-by-line, across the surface of a crystal. As the tip moves, it measures minute height differences in the crystal's surface due to the individual atoms. This is similar to the way sight-impaired people use their finger to sense braille on a page. The data from the tip is then sent to a computer that constructs an image of the atoms. An STM image of silicon atoms on the surface of a silicon wafer is shown in Figure 2.2.3.

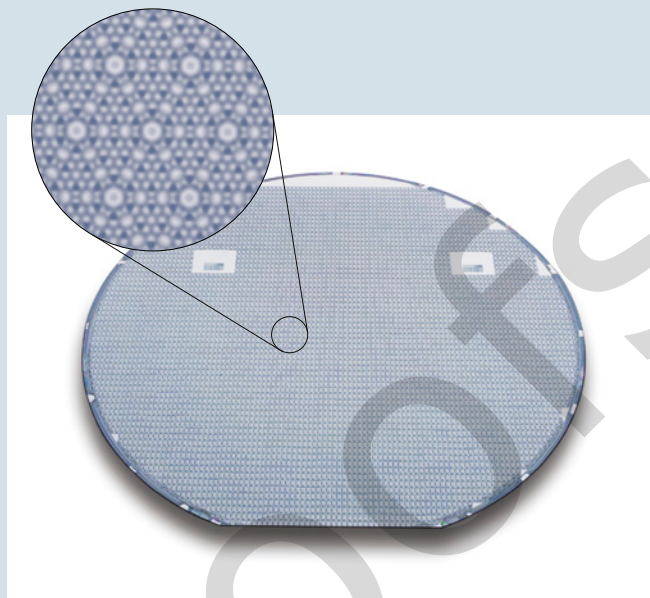


FIGURE 2.2.3 This image of the silicon atoms in a silicon wafer was taken with a scanning tunnelling microscope (STM).

EXTENSION

The periodic table: updated periodically

IUPAC (International Union of Pure and Applied Chemistry) is the worldwide representative body for chemists. In 2016 IUPAC announced the addition of four new elements to the periodic table. The four elements, which were discovered between 2004 to 2010, are:

- nihonium – symbol Nh, element 113
- moscovium – symbol Mc, element 115
- tennessine – symbol Ts, element 117
- oganesson – symbol Og, element 118.

Nihonium, tennessine and oganesson were prepared by bombarding previously known heavy atoms with other atoms, while moscovium is a radioactive decay product of tennessine.

Scientists are currently searching for the first element in the eighth row of the periodic table.

Atomic number

The number of protons in an atom's nucleus is known as the **atomic number** and is represented by the symbol Z .

All atoms that belong to the same element must have the same number of protons and therefore have the same atomic number, Z . For example, all hydrogen atoms have $Z = 1$, all carbon atoms have $Z = 6$ and all gold atoms have $Z = 79$.

Because all atoms are electrically neutral, the number of protons in an atom is always equal to the number of electrons in that atom. The atomic number therefore tells us both the number of protons and the number of electrons. For example, carbon atoms, with $Z = 6$, have six protons and six electrons.

Mass number

The number of protons and neutrons in the nucleus is known as the **mass number** and is represented by the symbol A . The mass number represents the total mass of the nucleus. Note that you cannot have fractions of a proton or neutron, therefore, the mass number is always a whole number.

The number of protons, neutrons and electrons defines the basic structure of an atom. The standard way of representing an atom is to show its atomic and mass numbers as shown in Figure 2.2.4.

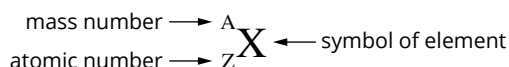


FIGURE 2.2.4 The standard way of representing an atom is to show its atomic number and mass number.

Worked example 2.2.1

CALCULATING THE NUMBER OF SUBATOMIC PARTICLES

Calculate the number of protons, neutrons and electrons for the atom with this atomic symbol: ${}^{40}_{18}\text{Ar}$	
Thinking	Working
The atomic number is equal to the number of protons.	The number of protons = $Z = 18$
Find the number of neutrons. Number of neutrons = mass number – atomic number	The number of neutrons = $A - Z$ = $40 - 18$ = 22
Find the number of electrons. The number of electrons is equal to the atomic number because the total negative charge is equal to the total positive charge.	Number of electrons = $Z = 18$

Worked example: Try yourself 2.2.1

CALCULATING THE NUMBER OF SUBATOMIC PARTICLES

Calculate the number of protons, neutrons and electrons for the atom with this atomic symbol:



2.2 Review

SUMMARY

- All the atoms of an element have the same number of protons. This number is called the atomic number (Z).
- In a neutral atom, the number of protons and electrons is equal.
- The mass number (A) of an atom is the total number of protons plus neutrons in the nucleus.

KEY QUESTIONS

- 1 Give the chemical symbol for the following elements.
 - a sodium
 - b copper
 - c neon
 - d potassium
 - e beryllium
 - f calcium
- 2 What term is given to the number of protons in the nucleus of an atom?
- 3 What term is given to the total number of protons and neutrons in the nucleus of an atom?
- 4 For any uncharged atom, which two subatomic particles are equal in number?
- 5 Calculate the numbers of protons, neutrons and electrons in the atom $^{31}_{15}\text{P}$.
- 6 What is the correct name of the element that has an atom with seven protons and eight neutrons?

2.3 Isotopes

One of the ideas Dalton put forward when he was developing modern atomic theory was that the atoms of an element are all identical; this is not strictly true.

All atoms that belong to the same element have the same number of protons in the nucleus and therefore the same atomic number, Z . However, not all atoms that belong to the same element have the same mass number, A . For example, hydrogen atoms can have mass numbers of 1, 2 or 3. In other words, hydrogen atoms may contain just a single proton, a proton and a neutron, or a proton and two neutrons as shown in Figure 2.3.1. Atoms that have the same number of protons (atomic number) but different numbers of neutrons (and therefore different mass numbers) are known as **isotopes**.

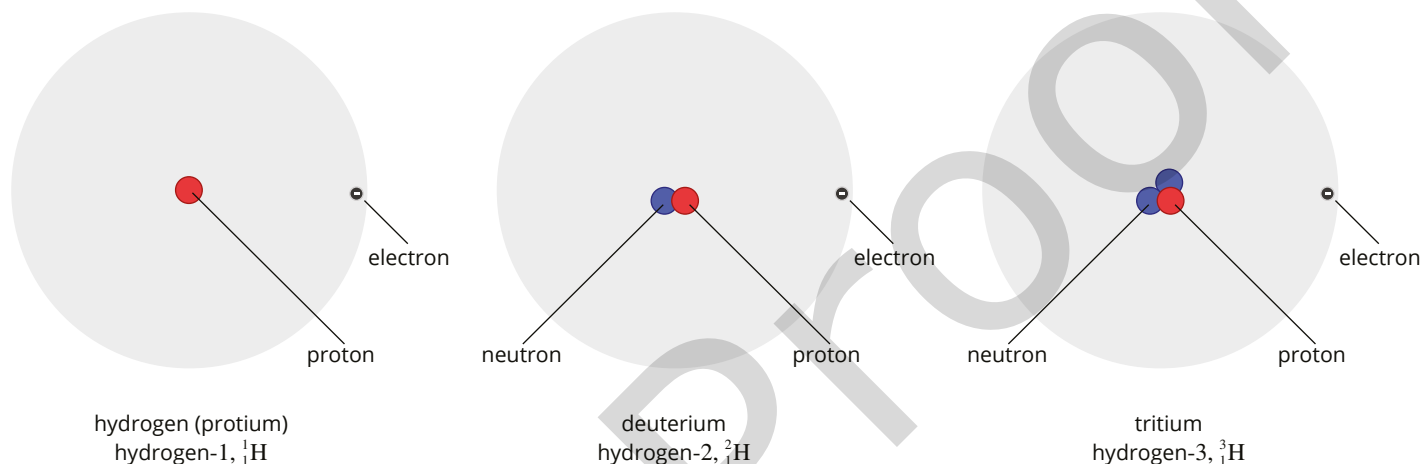


FIGURE 2.3.1 The three isotopes of hydrogen are given special names. A hydrogen atom with just one proton in its nucleus is known as hydrogen (or protium). A hydrogen atom with one proton and one neutron is known as deuterium. A hydrogen atom with one proton and two neutrons is known as tritium.

ISOTOPES

Carbon also has three naturally occurring isotopes. These three isotopes are known as carbon-12, carbon-13 and carbon-14. Carbon-12 atoms have a mass number of 12, carbon-13 atoms have a mass number of 13 and carbon-14 atoms have a mass number of 14. In the 1950s and 1960s, nuclear weapons testing caused a spike in carbon-14 in the atmosphere. This has been declining in the last 50 years. These three carbon isotopes can be represented as shown in Figure 2.3.2.

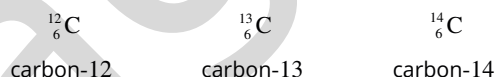


FIGURE 2.3.2 Carbon has three naturally occurring isotopes.

Isotopes have identical chemical properties but different physical properties such as mass and density. In particular, some isotopes that have a number of neutrons significantly different to the number of protons are unstable and hence **radioactive**. These isotopes undergo radioactive decay, emitting various forms of radiation and turn into lighter, more stable nuclei. Carbon-14, having 8 neutrons compared to 6 protons, is unstable and radioactive.

CHEMISTRY IN ACTION

Lucas Heights

Australia has 33% of the world's uranium deposits but no nuclear facilities generating electricity.

Australia does, however, have a working nuclear reactor in the Sydney suburb of Lucas Heights and has had since the late 1950s. The Open-Pool Australian Lightwater reactor (OPAL) is used for research, mineral analysis and the production of radioisotopes for medical and industrial applications (Figure 2.3.3).

In 1996 the residential area of Lucas Heights was renamed Barden Ridge as the association with the Lucas Heights nuclear reactor facility had adversely impacted the value of nearby real estate.

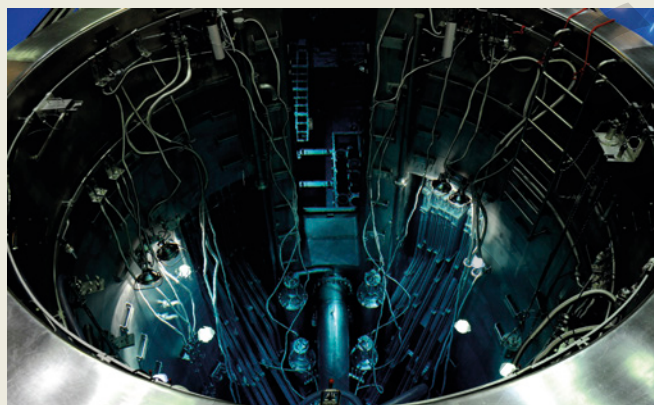


FIGURE 2.3.3 The OPAL nuclear research reactor at Lucas Heights in Sydney

Relative atomic masses

Chemists have devised a standard for measuring the masses of individual atoms. The standard chosen is the mass of an atom of the most common isotope of carbon, $^{12}_6\text{C}$ or carbon-12, which is given the value of 12 units. One unit of mass on this scale is therefore one twelfth of the mass of a $^{12}_6\text{C}$ isotope. The mass of an individual isotope of each element compared to this standard is known as its **relative isotopic mass**.

There are two different isotopes of the element chlorine; $^{35}_{17}\text{Cl}$ and $^{37}_{17}\text{Cl}$. The isotopes have different mass numbers because of the different numbers of neutrons. Remember that the mass of an atom is mainly determined by the number of protons and neutrons in their nucleus, since the mass of the electrons is relatively small.

The relative isotopic masses of the two chlorine isotopes are experimentally determined to be 34.969 (chlorine-35) and 36.966 (chlorine-37). Since the masses of a proton and a neutron are similar and close to 1 on the $^{12}_6\text{C} = 12$ scale, the relative isotopic mass of an isotope is almost, but not exactly, equal to the number of protons plus neutrons in the nucleus.

Table 2.3.1 below shows the relative isotopic mass and the relative abundances of the isotopes of some elements. The relative abundance of each isotope can be expressed as a percentage abundance and is used to indicate the proportion of each isotope in a sample of the element.

TABLE 2.3.1 The relative mass and abundance of hydrogen, oxygen and silver isotopes

Element	Isotopes	Relative isotopic mass	Relative isotopic abundance (%)
hydrogen	^1H	1.008	99.986
	^2H	2.014	0.014
	^3H	3.016	0.0001
oxygen	^{16}O	15.995	99.76
	^{17}O	16.999	0.04
	^{18}O	17.999	0.20
silver	^{107}Ag	106.9	51.8
	^{109}Ag	108.9	48.2

Relative isotopic masses do not have a unit because they are a comparison to the standard, carbon-12, which is 12 units exactly.

Most elements consist of a mixture of isotopes. For the purpose of the calculations you will be doing in later chapters, it is convenient to calculate the average relative mass of an atom in this naturally occurring mixture of isotopes. This average is known as the **relative atomic mass** of an element. It is sometimes referred to as atomic weight and given the symbol A_r .

The relative atomic mass of an element is a weighted average of the relative isotopic masses because it takes into account the relative abundances of each isotope in natural samples of the element.

One way of calculating the relative atomic mass of a sample of an element with a mixture of isotopes is to calculate the contribution of each isotope to the total mass, and then add these contributions. Another way is to use a mathematical expression to calculate the weighted mean. Both these methods are shown in Worked example 2.3.1.

i The relative atomic mass of an element is, the weighted average of the relative masses of the isotopes of the element relative to ^{12}C .

Worked example 2.3.1

CALCULATING THE RELATIVE ATOMIC MASS OF CHLORINE: METHOD 1

Chlorine is made up of two isotopes, ^{35}Cl and ^{37}Cl . ^{35}Cl has a relative isotopic mass of 34.969 and a relative abundance of 75.80%. ^{37}Cl has a relative isotopic mass of 36.966 and a relative abundance of 24.20%. Calculate the relative atomic mass of chlorine.	
Thinking	Working
Calculate the relative atomic mass contribution by the chlorine-35 isotope, which is the relative isotopic mass of the Cl-35 isotope, times the percentage abundance, divided by 100.	$34.969 \times \frac{75.80}{100} = 26.50$
Calculate the relative atomic mass contribution by the chlorine-37 isotope.	$36.966 \times \frac{24.20}{100} = 8.945$
Add the relative atomic mass contributions of all the isotopes.	$26.50 + 8.945 = 35.45$

Worked example 2.3.1

CALCULATING THE RELATIVE ATOMIC MASS OF CHLORINE: METHOD 2

Chlorine is made up of two isotopes, ^{35}Cl and ^{37}Cl . ^{35}Cl has a relative isotopic mass of 34.969 and a relative abundance of 75.80%. ^{37}Cl has a relative isotopic mass of 36.966 and a relative abundance of 24.20%. Calculate the relative atomic mass of chlorine.	
Thinking	Working
Identify the relative isotopic masses and the abundance of each isotope.	^{35}Cl relative isotopic mass = 34.969 relative abundance = 75.80% ^{37}Cl relative isotopic mass = 36.966 relative abundance = 24.20%
Use the expression for calculating the weighted mean (A_r) from the percentage abundance and the relative isotopic masses: $A_r = \frac{(\% \times \text{relative isotopic mass}) + (\% \times \text{relative isotopic mass})}{100}$	$A_r = \frac{(75.80 \times 34.969) + (24.20 \times 36.966)}{100}$
Calculate the relative atomic mass. (Give answer to 4 significant figures.)	$A_r = 35.45$

Worked example: Try yourself 2.3.1

CALCULATING THE RELATIVE ATOMIC MASS OF SILVER

Use the data from Table 2.3.1 to calculate the relative atomic mass of silver.

If you are given the relative atomic mass of a sample of an element, and you know that only two isotopes are present, it is also possible to work backwards and calculate the relative abundances of the isotopes. The technique for doing this is shown in Worked example 2.3.2.

Worked example: 2.3.2

CALCULATING THE ABUNDANCE OF ISOTOPES FROM RELATIVE ATOMIC MASSES AND ISOTOPIC MASSES

The relative atomic mass of rubidium is 85.47. The relative isotopic masses of its two isotopes, ^{85}Rb and ^{87}Rb , are 84.95 and 86.94. Calculate the relative abundances of the isotopes.	
Thinking	Working
Use a to represent the percentage abundance of the lighter isotope. The percentage abundance of the heavier isotope must therefore equal $(100 - a)$.	Abundance of $^{85}\text{Rb} = a$ Abundance of $^{87}\text{Rb} = (100 - a)$
Substitute the values into the following expression (used in method 2 above): $A_r = \frac{(\% \times \text{relative isotopic mass}) + (\% \times \text{relative isotopic mass})}{100}$	$85.47 = \frac{(84.95 \times a) + (86.94 \times (100 - a))}{100}$
Expand the top line of the equation.	$85.47 = \frac{84.95a + 8694 - 86.94a}{100}$
Solve the equation to find a (the abundance of the lighter isotope).	$8547 = 84.95a + 8694 - 86.94a$ $8547 - 8694 = 84.95a - 86.94a$ $-147 = -1.99a$ $a = \frac{147}{1.99} = 73.87$ Abundance of $^{85}\text{Rb} = 73.87\%$
Determine the abundance of the heavier isotope using $(100 - a)$.	Abundance of $^{87}\text{Rb} = 100 - 73.87$ $= 26.13\%$

CHEMFILE

New standard atomic weights for 19 elements

In 2013 IUPAC voted to change the standard atomic weights of 19 elements. Of the 19 elements, four of them have had renewed isotopic abundance values leading to a change in standard atomic weights. For the remaining 15, the new weights are a result of improvements to the precision of analytical devices.

With improvements to the accuracy of these values, all other sciences and research which involves these values will also see an improvement in accuracy.

Worked example: Try yourself 2.3.2

CALCULATING THE ABUNDANCE OF ISOTOPES FROM RELATIVE ATOMIC MASSES AND ISOTOPIC MASSES

The relative atomic mass of copper is 63.54. The relative isotopic masses of its two isotopes, ^{63}Cu and ^{65}Cu , are 62.95 and 64.95. Calculate the relative abundances of the isotopes.

For comparison purposes, the standard atomic weight can also be used. The **standard atomic weight** is the relative atomic mass of the element based on the agreed proportions of isotopes in a 'normal' sample of the element on Earth.

2.3 Review

SUMMARY

- The isotopes of an element have the same atomic number but different mass numbers. They have the same number of protons but different numbers of neutrons.
- Isotopes of an element have the same chemical properties but different physical properties.
- Isotopes with significantly different number of neutrons to protons tend to be unstable and therefore radioactive.
- Radioactive isotopes undergo radioactive decay, emitting various forms of radiation to become lighter, more stable nuclei.
- Relative atomic masses are the weighted average of the relative isotopic masses taking into account their relative abundance in nature.

KEY QUESTIONS

- 1 What is an isotope?
- 2 Explain the similarities and differences between isotopes of the same element.
- 3 How many more neutrons does an atom of carbon-14 have than an atom of the carbon-12 isotope?
- 4 Use the data in Table 2.3.1 on page 12 to calculate the relative atomic mass of:
 - a oxygen
 - b hydrogen.
- 5 The element lithium has two isotopes:
 - ${}^6\text{Li}$ with a relative isotopic mass of 6.02
 - ${}^7\text{Li}$ with a relative isotopic mass of 7.02The relative atomic mass of lithium is 6.94. Calculate the percentage abundance of the lighter isotope.

2.4 Mass spectrometry

Mass spectrometry is a technique used to measure the mass of atoms or molecules. In this context of atomic structure, it can be used to identify the presence and relative abundance of isotopes in a sample of an element. In other situations, it can also be used to measure the mass of molecules, which provides evidence to help determine the structure of the molecules.

A MASS SPECTROMETER

Relative isotopic masses of elements and their isotopic abundances are determined by using an instrument called a **mass spectrometer**, which was invented by Francis Aston in 1919. This instrument:

- separates the individual isotopes in a sample of the element
- determines the mass of each isotope, relative to the carbon-12 standard
- calculates the relative abundances of the isotopes in the sample.

A mass spectrometer, as shown in Figure 2.4.1, operates as follows:

- 1 The sample is vaporised and then ionised using high energy electrons.
- 2 The ions are separated and accelerated according to their mass-to-charge $\frac{m}{z}$ ratios in an electric/magnetic field.
- 3 The ions that have a particular mass-to-charge ratio are detected by a device that counts the number of ions that strike it.

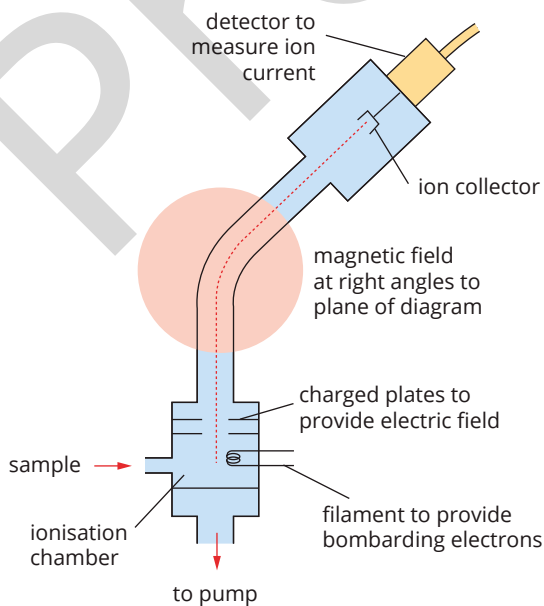


FIGURE 2.4.1 A mass spectrometer is an apparatus used for separating particles, such as isotopes, according to their mass and charge.

Mass spectra

The information obtained from a mass spectrometer is presented graphically as a **mass spectrum**, which plots the relative abundance of each ion against their mass-to-charge ratio, $\frac{m}{z}$.

For simple mass spectra, most of the ions produced in the instrument have a +1 charge, so the value of the $\frac{m}{z}$ ratio is equal to the mass of the ion.

A mass spectrum of aluminium is shown in Figure 2.4.2. The isotopic composition of an element can be seen from its mass spectrum. Aluminium exists as only one natural isotope and so there is only one peak (line) in the mass spectrum of this element. The mass-to-charge ratio corresponds with the aluminium's relative isotopic mass.

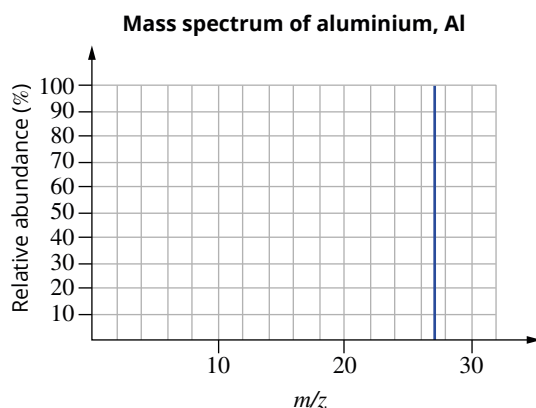


FIGURE 2.4.2 The mass spectrum of aluminium. There is only one peak because aluminium consists of only one isotope.

Many elements occur as a mixture of two or more isotopes. For most elements, the proportion of its isotope is approximately the same in every sample, regardless of from where on Earth the sample has been taken. The percentage amount of an isotope in the natural element is called the relative isotopic abundance of the isotope in that element.

The mass spectrum of magnesium is shown in Figure 2.4.3. Magnesium consists of three isotopes, $^{24}_{12}\text{Mg}$, $^{25}_{12}\text{Mg}$ and $^{26}_{12}\text{Mg}$ with relative abundances of 79, 10 and 11%, respectively, and so its mass spectrum contains three peaks with the height of each peak representing 79%, 10% and 11% respectively of the total peak height.

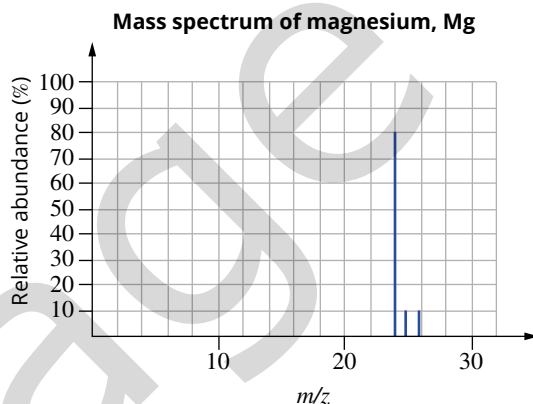


FIGURE 2.4.3 A mass spectrum of magnesium showing three peaks for its three different isotopes

Worked example 2.4.1

CALCULATING PERCENTAGE ABUNDANCE OF EACH ISOTOPE FROM THE MASS SPECTRUM

Figure 2.4.4 shows a simplified mass spectrum of magnesium. From this mass spectrum, calculate the percentage abundance of each of its three isotopes.

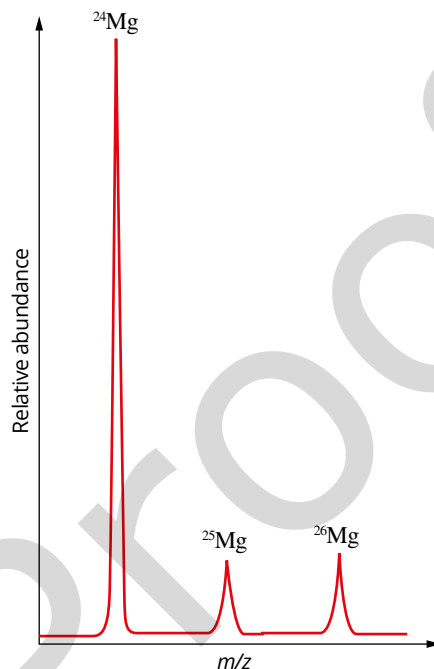


FIGURE 2.4.4 Mass spectrum of magnesium

Thinking	Working
Measure the peak height for each isotope using a ruler.	From the spectrum, the height of each peak is: $^{24}\text{Mg} = 7.9 \text{ cm}$ $^{25}\text{Mg} = 1.0 \text{ cm}$ $^{26}\text{Mg} = 1.1 \text{ cm}$
Calculate the total peak height for the three isotopes by adding the individual peak heights.	Total peak height = $7.9 + 1.0 + 1.1$ $= 10.0 \text{ cm}$
Substitute the peak height for each isotope into the formula: $\% \text{ abundance} = \frac{\text{peak height}}{\text{total peak height}} \times 100\%$	$\% \text{ abundance } ^{24}\text{Mg} = \frac{7.9}{10.0} \times 100$ $= 79\%$ $\% \text{ abundance } ^{25}\text{Mg} = \frac{1.0}{10.0} \times 100$ $= 10\%$ $\% \text{ abundance } ^{26}\text{Mg} = \frac{1.1}{10.0} \times 100$ $= 11\%$

Worked example: Try yourself 2.4.1

CALCULATING PERCENTAGE ABUNDANCE OF EACH ISOTOPE FROM THE MASS SPECTRUM

Figure 2.4.5 shows a simplified mass spectrum of lead. From this mass spectrum, calculate the percentage abundance of each of its three isotopes.

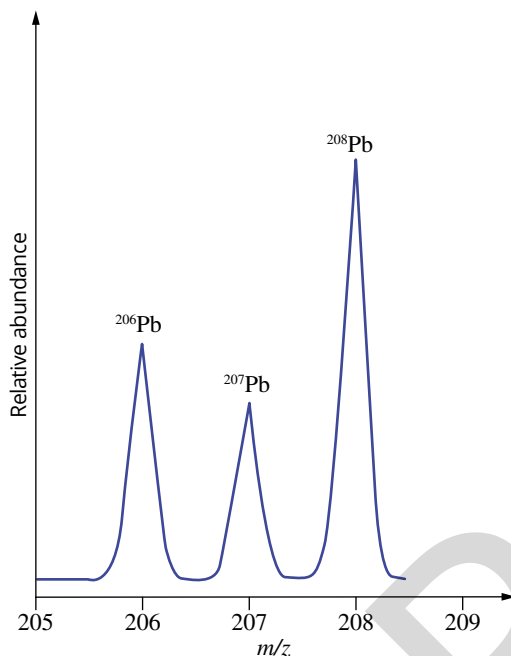


FIGURE 2.4.5 Mass spectrum of lead

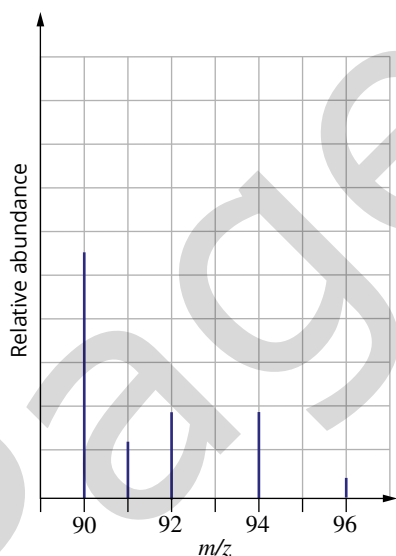
2.4 Review

SUMMARY

- Isotopic abundances can be measured by mass spectrometry.
- A mass spectrometer:
 - separates the individual isotopes in a sample of the element
 - determines the mass of each isotope, relative to the carbon-12 standard
 - calculates the relative abundances of the isotopes in the sample.
- A mass spectrum is a plot of the measured abundance of each isotope against the mass-to-charge ratio.
- In a mass spectrum of an element, the:
 - number of peaks indicates the number of isotopes
 - position of each peak on the horizontal axis indicates the relative isotopic mass
 - relative heights of the peaks correspond to the relative abundance of the isotopes.

KEY QUESTIONS

- What are the three key operating steps of a mass spectrometer?
- What does the number of peaks in a mass spectrum represent?
- What does the position of the peaks along the x-axis represent?
- What does the height of peaks represent?
- The mass spectrum of zirconium is shown in the graph below.
 - Measure the peak heights to calculate the percentage abundance of each zirconium isotope.
 - Use the percentage abundances calculated in part **a** to determine the relative atomic mass of zirconium.



Chapter review

KEY TERMS

atomic number
chemical symbol
electron
electrostatic attraction
isotope
mass number

mass spectrometer
mass spectrum
neutron
nucleon
nucleus
periodic table

proton
radioactive
relative atomic mass
relative isotopic mass
standard atomic weight

Atomic theory

- 1 Which of Dalton's predictions about the nature of atoms was later proven to be incorrect?
- 2 Where would you find 99.97% of the mass of an atom?
- 3 How are protons, neutrons and electrons arranged in an atom?
- 4 Compare the mass and charge of protons, neutrons and electrons.
- 5 Who is credited with discovering each of the subatomic particles?

Describing atoms

- 6 Is the periodic table organised by atomic number or mass number or both?
- 7 An atom of chromium can be represented by the symbol $^{52}_{24}\text{Cr}$.
 - a Determine its atomic number and mass number.
 - b Determine the number of electrons, protons and neutrons in the chromium atom.
- 8 An atom of manganese can be represented by the symbol $^{55}_{25}\text{Mn}$.
 - a Determine its atomic number and mass number.
 - b Determine the number of electrons, protons and neutrons in the manganese atom.
- 9 Explain why the number of electrons in an atom equals the number of protons.
- 10 Using the element bromine as an example, explain why elements are best identified by their atomic number and not their mass number.

Isotopes

- 11 Two atoms both have 20 neutrons in their nucleus. The first has 19 protons and the other has 20 protons. Are they isotopes? Why or why not?

- 12 The standard on which all relative masses are based is the ^{12}C isotope, which is given a mass of 12 exactly. Explain why, in the table of relative atomic masses in the Appendix at the end of the book, the relative atomic mass of carbon is listed as 12.01.
- 13 Determine the percentage abundance of the lighter isotope of each of the following elements.
 - a Gallium: relative isotopic masses 68.95 and 70.95 respectively; $A_r = 69.72$
 - b Boron: relative isotopic masses 10.02 and 11.01 respectively; $A_r = 10.81$
- 14 The separation of certain isotopes from the bulk of an element is a highly technical process. Why can this process not be carried out by chemical means?
- 15 The two isotopes argon-40 and calcium-40 both have the same mass number. In terms of the numbers of protons, neutrons and electrons, what do these two isotopes share the same number of?

Mass spectrometry

- 16 When a sample of palladium is placed in a mass spectrometer, the peaks are recorded as the relative isotopic masses and corresponding percentage abundances given in the table below.

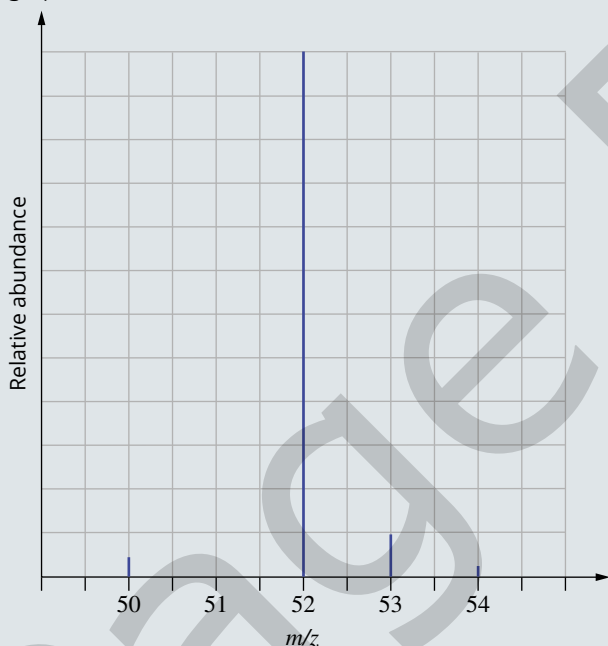
Relative isotopic mass	Abundance (%)
101.9049	0.9600
103.9036	10.97
104.9046	22.23
105.9032	27.33
107.9039	26.71
109.9044	11.80

Calculate the relative atomic mass of palladium.

- 17 The table below gives isotopic composition data for argon and potassium.

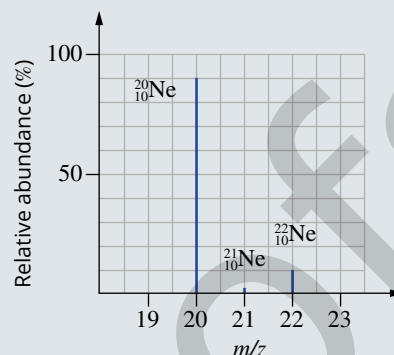
Element	Atomic number	Relative isotopic mass	Relative abundance (%)
argon	18	35.978	0.307
		37.974	0.060
		39.974	99.633
potassium	19	38.975	93.3
		39.976	0.011
		40.974	6.69

- a Determine the relative atomic masses of argon and potassium.
 b Explain why the relative atomic mass of argon is greater than that of potassium, even though potassium has a larger atomic number.
- 18 The mass spectrum of chromium is shown in the graph below.



- a Measure the peak heights to calculate the percentage abundance of each chromium isotope.
 b Use the percentage abundances calculated in part a to determine the relative atomic mass for chromium.
- 19 The relative atomic mass of europium is 151.96. The relative isotopic masses of its two isotopes are 150.92 and 152.92. Calculate the relative abundances of the isotopes in naturally occurring europium.

- 20 Based on the mass spectrum of neon shown below, give an estimate of the relative atomic mass of neon to the closest whole number.



Connecting the main ideas

- 21 Using suitable examples, clearly distinguish between:
 a mass number
 b relative isotopic mass
 c relative atomic mass.
- 22 The Banana Equivalent Dose is an informal expression of ionising radiation exposure a person would be exposed to by eating one average-sized banana. The major source of radioactivity in bananas is the naturally occurring radioactive isotope potassium-40. A sample of bananas was homogenised, dissolved and the potassium extracted. The resultant solution was analysed by mass spectrometry producing the following results.

Relative isotopic mass	Peak area
38.96	454 212.6
39.96	56.98473
40.96	32 779.37

- a Calculate the percentage abundance of the radioactive potassium-40 isotope.
 b Using the results, determine the relative atomic mass of potassium to 2 decimal places.
- 23 Suppose an element has three isotopes. The first isotope has a mass number of x with a relative abundance of 35%. The second isotope has a mass number of $x + 2$ and a relative abundance of 54%, and the third isotope has a mass number of $x + 3$.
 a If the peak of the first isotope of the mass spectrum is 25 mm long, calculate the height of the second peak. Give your answer to the nearest whole number.
 b Calculate the height of the third peak (give your answer as a whole number).